



MTH101, Basic Linear Algebra

Monsoon 2024

29/08/2024, 2:00PM-3:00PM.

Tutorial-1

(1) Draw the solutions of the following equations¹ in $\mathbb{R}^2$.

a $2X+Y=2$

b $Y=0$

c $X-Y=5$

(2) Find all the solutions of the system of equations

$$X = 5$$

$$Y = 5$$

$$X - Y = 0$$

(3) Let $P_1 = (a, b) \in \mathbb{R}^2$ and $P_2 = (c, d) \in \mathbb{R}^2$. Find a linear equation for which P_1 and P_2 are solutions. How many equations are there having P_1 and P_2 as solutions?

(4) Find all the linear equations having $(0, 0) \in \mathbb{R}^2$ as one of its solutions. Justify your answer!

(5) For distinct numbers $a, b \in \mathbb{R}$, find the linear equation for which $(5, a)$ and $(5, b) \in \mathbb{R}^2$ are solutions.

(6) Draw the solutions of the equation

$$Y - X = 2.$$

Write down some solutions $(p, q) \in \mathbb{N}^2$, where $\mathbb{N} = \{1, 2, 3, \dots\}$, such that both p and q are prime numbers²

¹If there is only one variable present in the equation then treat the coefficient of the other one to be zero.

²It is not known whether there are infinitely many such pairs! Search on internet "twin prime conjecture".



Tutorial-2

(1) Find 2×2 nonzero matrices A_1, A_2 and A_3 of your choice such that $A_1^2 = 0$, $A_2^2 = I_2$ and $A_3^3 = I_2$.

(2) Think of points $(x, y) \in \mathbb{R}^2$ as 2×1 matrices

$$X := \begin{pmatrix} x \\ y \end{pmatrix}$$

and calculate the set $L := \{AX \mid \text{for any } X \in \mathbb{R}^2\}$ for the following matrices

(a) $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

(b) $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$

(c) $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$

(d) $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

In each of the above cases, geometrically describe¹ the set L .

(3) Verify that the set of all 2×2 matrices satisfies the associativity property, that is, for any three 2×2 matrices A, B and C ,

$$(AB)C = A(BC).$$

(4) Let L be the line in $\mathbb{R}^2$ having distinct points

$$\begin{pmatrix} a \\ b \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} c \\ d \end{pmatrix},$$

where $a \neq 0$. For an invertible matrix A , show that the set of all points

$$\{AX \mid X \text{ is a point on the line } L\}$$

form a line² in $\mathbb{R}^2$.

¹Is it a point or a line or the whole of $\mathbb{R}^2$.

²Produce the equation of the line!



Tutorial-3

(1) Which of the following matrices are in reduced row echelon form?

(a) $\begin{pmatrix} 0 & 0 & 1 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 0 & 0 & 1 & 0 & 5 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$

(2) Solve the system of equations by row reducing the corresponding augmented matrix to its echelon form.

(a)

$$\begin{aligned} X + 2Y + Z &= 1 \\ X - Y + 2Z &= -2 \\ X + 3Y + -Z &= 5 \end{aligned}$$

(b)

$$\begin{aligned} 2X + 3Y + Z + 4W &= 1 \\ X + 4Y + 2Z + W &= 0 \\ Z - W &= 0 \end{aligned}$$

(3) Let E_{ij} denote the 4×4 matrix whose ij^{th} entry is 1 and 0 everywhere else.

(a) Calculate the product $E_{21}E_{13}$

(b) Calculate the product $E_{31}E_{23}$

(4) Let

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{pmatrix}.$$

Verify the following

(a) The row operation $R_4 \rightarrow R_4 + -4R_1$ on A is the same as premultiplying A with $I_4 + -4E_{41}$.

(b) Find the 4×4 matrix B so that BA is in row reduced echelon form.



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Tutorial-4

- (1) Find the inverse of the following matrices, if they exist, by transforming them to their row reduced echelon form.

(a) $\begin{pmatrix} 3 & 0 & 1 \\ 1 & 4 & 3 \\ 4 & 1 & 2 \end{pmatrix}$

(b) $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$

- (2) Let I_n be the $n \times n$ identity matrix and A be a matrix such that $A^k = 0$ for some positive integer $k \geq 1$. Prove that $I_n + A$ is an invertible matrix.

- (3) Using the principles of mathematical induction, verify the formula

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}, \quad \text{for any natural number } n.$$

(4) Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, $B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$ and $C = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{pmatrix}$.

Show that

$$\det(C) = \det(A) + \det(B).$$

¹A matrix A such that $A^k = 0$ for some positive integer k is called a *nilpotent* matrix.



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Tutorial-5

- (1) Let A be an $n \times n$ matrix. Using the principle of mathematical induction, verify that if A has a zero row then $\det(A) = 0$.
- (2) Let A be an $n \times n$ matrix. Using the principle of mathematical induction, verify that if A has a zero column then $\det(A) = 0$.
- (3) Let Y be a point in $\mathbb{R}^n$ (seen as a $n \times 1$ matrix) and A be any invertible $n \times n$ matrix. Show that there is a point X such that $AX = Y$.
- (4) An $n \times n$ matrix A is called upper triangular if $a_{ij} = 0$ for $1 \leq j < i \leq n$. Using the principle of mathematical induction, show that $\det(A) = a_{11}a_{22} \cdots a_{nn}$.
- (5) Let E be a $n \times n$ elementary matrix and A be any $n \times n$ matrix. Then verify that $\det(EA) = \det(E) \det(A)$.



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Midterm Examination

(1) (5pts) Find the inverse of the matrix

$$U = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

where a is a real number, by finding the reduced row echelon form of U .

(2) (5pts) Let A and B be invertible $n \times n$ matrices. Verify that

(i) $\det(A^{-1}) = \frac{1}{\det(A)}$

(ii) $\det(B^{-1}AB) = \det(A)$.

(3) (5pts) Let $\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Find the 2×2 matrix A so that $A\mathbf{x}_1 = \mathbf{x}_2$ and $A\mathbf{x}_2 = \mathbf{x}_1$.

(4) (5pts) Let A be an $n \times n$ matrix and $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n$ be a point, which is viewed as an $n \times 1$ matrix, such that $A\mathbf{x} = \mathbf{0}$. Prove that $\det(A) = 0$.

(5) (5pts) Find the set of all solutions of the following system of equations

$$X + 6Y + 4Z + 2W = 0$$

$$X + 3Y + Z - W = 1$$

by reducing the corresponding augmented matrix to its reduced row echelon form.



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Tutorial-7

Throughout, V is an F -vector space.

- (1) Let W be a subspace of V and S be a nonempty subset of W . Prove that $\text{span}(S) \subseteq W$.
- (2) Let W_1 and W_2 be two subspaces of V . Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
- (3) Let W_1 and W_2 be two subspaces of V . Prove that
 - (a) $W_1 + W_2$ is a subspace of V .
 - (b) the smallest subspace containing $W_1 \cup W_2$ is the subspace $W_1 + W_2$.
- (4) Prove that any three vectors in $\mathbb{R}^2$ are linearly dependent.

(5) Let

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}.$$

Prove that W is a finite dimensional subspace of $\mathbb{R}^3$ by explicitly producing two vectors $w_1, w_2 \in W$ such that $\text{span}\{w_1, w_2\} = W$.

- (6) Let $v_1, \dots, v_n$ be linearly independent vectors in V and $v \in V$. Prove that $v, v_1, \dots, v_n$ are linearly independent if and only if $v \notin \text{span}\{v_1, v_2, \dots, v_n\}$.

(7) Let

$$\mathcal{P} := \{a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 : a_i \in \mathbb{R}, n \in \mathbb{N}\}$$

be the set of all polynomials over real numbers.

(a) Prove that $\mathcal{P}$ is an $\mathbb{R}$ -vector space under the following operation

vector space addition:
$$\sum_{i=0}^n a_i x^i + \sum_{j=0}^m b_j x^j = a_0 + b_0 + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots$$

scalar multiplication:
$$\alpha \sum_{i=0}^n a_i x^i = \sum_{i=0}^n (\alpha a_i) x^i.$$

(b) Prove that $\mathcal{P}$ is not a finite dimensional vector space.

- (8) Let $0 \neq v \in V$ and $u \in V$ be an element that is not on the line joining 0 and v . Suppose that $T_1 : V \rightarrow V$ and $T_2 : V \rightarrow V$ are linear transformations such that $T_1(v) = T_2(v)$ and $T_1(u) = T_2(u)$. Prove that T_1 and T_2 are the same linear transformation. That is, $T_1(x) = T_2(x)$ for every element $x \in V$.



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Tutorial-8

- (1) Using results proved in previous lectures on injective linear transformations, explain why a homogeneous system of m equations and n unknowns have infinitely many solutions when $n > m$.
- (2) Let $T : V \rightarrow W$ be a one to one linear transformation of finite dimensional vector spaces. If $\text{Dim}_F(V) = \text{Dim}_F(W)$ then prove that T is an isomorphism.
- (3) Let $T : V \rightarrow W$ be an onto linear transformation of finite dimensional vector spaces. If $\text{Dim}_F(V) = \text{Dim}_F(W)$ then prove that T is an isomorphism.
- (4) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$ be a nonzero linear transformation of $\mathbb{R}$ -vector spaces, where $n \geq 2$. Find the dimension of $\ker(T)$.
- (5) For any two linear transformations $T_1 : V \rightarrow W$ and $T_2 : V \rightarrow W$ of F -vector spaces, define

$$T_1 + T_2 (v) = T_1(v) + T_2(v)$$

$$\alpha \star T_1 (v) = \alpha T_1(v).$$

Show $T_1 + T_2$ and $\alpha \star T_1$ are linear transformations from V to W and that the collection $\mathcal{L}(V, W)$ of all linear transformations from V to W forms an F -vector space under the above addition and scalar multiplication operations.



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Tutorial-9

Definition: For an $n \times n$ matrix A , the polynomial $\det(A - \lambda I_n)$ is called the *characteristic polynomial* of A and its roots are called the *eigenvalues* of A .

- (1) Let A be an $n \times n$ matrix with real number entries and $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear transformation defined by $Tx := Ax$ for all $x \in \mathbb{R}^n$. Calculate the dimension of $\ker T$ and $\operatorname{im} T$ for the following choices of A .

(i) $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

(ii) $A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

- (2) Find the eigenvalues and eigenvectors of the following matrices

(i) $\begin{pmatrix} -7 & -2 & 10 \\ -3 & 2 & 3 \\ -6 & -2 & 9 \end{pmatrix}$

(ii) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix}$

- (3) For any $n \times n$ matrix A , show that A and A^T have the same characteristic polynomial (in particular, they have the same eigenvalues).
- (4) If $v \in \mathbb{R}^n$ is an eigenvector of an $n \times n$ matrix A then v is also an eigenvector of A^m for every positive integer m .
- (5) If A is an $n \times n$ matrix such that $\det A = 0$ then prove that 0 is an eigenvalue of A and furthermore, if $A^m = 0$ for some positive integer m then prove that 0 is the only eigenvalue of A .